

Rafik El Khoury

Urban Planner · AI & Computational Design

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Profile

Urbanist who loves cities and the built environment, with degrees in civil engineering, planning and urban design, AI for architecture and business innovation. I plan municipal and heritage sites and build the AI tools my firm's architects use. At ease with mayors, clients and site teams; quick to deliver, careful with numbers. Seeking planning, urban design or AI roles.

Experience

AI Implementation Strategist / Urban Planner, Rafik El-Khoury & Partners – Lebanon / UAE Aug 2026 – present

- Reframed a municipality's VR-museum and training-hub brief for a 12th-century castle in Akkar into a phased heritage-gateway concept, once GIS and 10-year climate analysis (87,672 hourly ERA5 records) ruled out the original programme.
- Gave the office's architects AI agents inside Revit by hardening an open-source Revit MCP server they now use (token auth, fail-closed writes, 9 new tools, tests up from 64 to 1,240); added human-approved OmniClass classification, scoped for ~400 models.
- Built Render App, now being adopted by the architects: SketchUp views, sketches and plans become client-ready renders that keep the drawn geometry (depth ControlNet, 18 tasks, spend cap, 300+ tests; Python, fal.ai, React).
- Piloting an AI-agent layer on the firm's ERPNext (scoped as a 56-agent catalogue) that plugs in the Revit server and my Teams/Telegram timesheet bot, which logs plain-language hours in staff workbooks (329 staff, 418 projects, 703 tests).
- Built C-Gate, a deterministic, citation-grounded compliance checker for international procurement tenders that flags issues and leaves decisions to people. Rebuilt the company website in Astro around a 528-project archive (in preview).

Graduate Urban Planner, Rafik El-Khoury & Partners – Abu Dhabi, UAE Jan 2025 – Sept 2025

- Scored international masterplan proposals for a 19 km² island development with a framework weighting design merit, consultant expertise and fit with the long-term vision.
- Worked on the SHA villa complex redesign in Al-Jurf: peer-reviewed piling works, refined the CAD set and designed the sales brochure.
- Programmed three community parks in Madinat Al-Riyadh from site analysis, and wrote school-rehabilitation proposals for southern Lebanon.
- Briefed the president of Aramoun municipality on the town's urban character, planning issues and proposed interventions.

Site Engineer, IMKAN Properties – UAE Jan 2023 – Mar 2023

- Inspected villas at every construction stage, waterproofing to tiling, and issued weekly progress reports.

Selected Projects

CoolSpend, 3rd place at the Infrared.city Buildathon 2026 (solo) 2026

Barcelona street-tree planner: fuses four open datasets into a 494-cell heat-vulnerability grid and spends a fixed budget where heat hurts most, measured on live UICI runs. €1M buys 90 trees, 20,609 m² cooled, 26,745 residents served.

FestCOOL / U-Shade, IAAC (team of 4) 2026

Places festival shade where heat and crowds overlap (UICI grid × JuPedSim crowds, maximal covering), tested on the Primavera Sound 2026 site.

ARCHAI, IAAC (team of 4) 2025–2026

Checks Rhino building geometry against zoning rules and explains each pass or fail with the cited article (retrieval over Mumbai and Barcelona codes).

GigAI, IAAC (team of 3) 2026

Turns site-meeting transcripts into confidence-scored material-change actions for PM approval (RAG, pgvector, Claude).

Education

IAAC and EADA Business School, Barcelona, Master in AI for Architecture and Business Oct 2025 – July 2026
Innovation (double degree, IAAC + EADA)

University of Dundee, MSc in Spatial Planning with Sustainable Urban Design Sept 2023 – Sept 2024

Loughborough University, BEng in Civil Engineering (dissertation: flood management) Oct 2018 – June 2022

Skills

Planning & design: Masterplan evaluation, site analysis, heritage reuse, public space, AutoCAD, Revit, Rhino, Blender

Spatial & data: GIS (QGIS, OSM, SRTM), ERA5 climate data, UICI simulation (Infrared.city), Python, optimisation

AI & automation: Multi-agent research pipelines, MCP servers, RAG, local LLMs (Ollama), FastAPI, React, Docker

Languages: Arabic (native), English (fluent), French (fluent), Spanish (conversational)